

Data Science and Analytics

2. Statistical Foundation

Er. Narayan Sapkota, M.Sc.

Everest Engineering College

narayan.sapkota@eemc.edu.np

May 9, 2026

Table of Contents

1 Types of Variables: Numeric and Categorical

2 Empirical Distribution

- Numeric Data Distributions
- Categorical Data Distributions

3 Correlation Analysis

- Pearson Correlation
- Cross-tabulation
- Numeric and Categorical Correlation

4 Statistical Significance

- p-value
- t-value
- Chi-Squared Test

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
- 3 Correlation Analysis
- 4 Statistical Significance

Types of Variables (1)

Variables

- A variable is a measurable characteristic or attribute that can take different values.
- Understanding variable types is essential for selecting appropriate statistical and machine learning techniques.

1. Numeric Variables:

- Represent measurable quantities expressed as numbers.
- Can be used for arithmetic operations such as addition, averaging, and comparison.
- Two main types:
 - **Discrete:** Countable values (e.g., number of students)
 - **Continuous:** Infinite values within a range (e.g., height, weight)
- **Examples:**
 - Age of a person
 - Salary of employees
 - Temperature readings
 - Number of website visits

Types of Variables (2)

2. Categorical Variables:

- Represent qualitative data divided into categories or groups.
- Cannot be meaningfully used in arithmetic operations.
- Two main types:
 - **Nominal:** No natural order (e.g., colors, gender)
 - **Ordinal:** Ordered categories (e.g., low, medium, high)
- **Examples:**
 - Gender (Male/Female/Other)
 - Blood group
 - Product category
 - Education level
- **Numeric variables measure quantity, while categorical variables describe groups or labels.**

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
 - Numeric Data Distributions
 - Categorical Data Distributions
- 3 Correlation Analysis
- 4 Statistical Significance

Empirical Distribution (1)

Empirical Distribution

- A statistical representation of observed data based on actual samples collected from a population.
- It describes how values are distributed in a dataset without assuming any predefined mathematical model.
- It is constructed directly from data observations rather than theoretical assumptions.
- Each observed value contributes to the overall shape of the distribution.

Key Idea

- The empirical distribution approximates the true (but unknown) distribution of the population.
- As sample size increases, it becomes a better representation of the underlying reality.

Types of Empirical Representation:

Empirical Distribution (2)

- **Empirical Frequency Distribution:** Counts how often each value or range occurs.
- **Empirical Cumulative Distribution Function (ECDF):** Shows cumulative probability of observed values.
- **Relative Frequency Distribution:** Represents proportions instead of raw counts.

```
data = np.array([10, 12, 12, 15, 18, 18, 18, 20, 22, 25])

# -----
# 1. Frequency Distribution
# -----
values, counts = np.unique(data, return_counts=True)

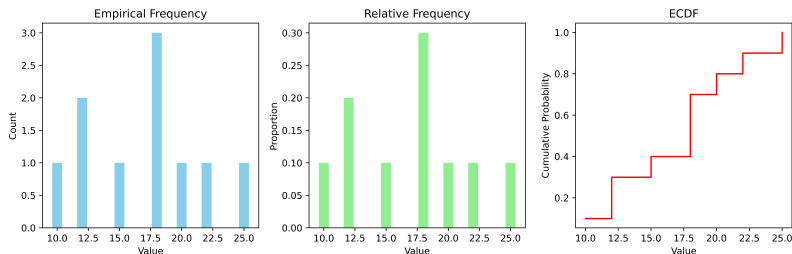
# -----
# 2. Relative Frequency
# -----

rel_freq = counts / len(data)

# -----
# 3. ECDF
```


Empirical Distribution (3)

```
# -----  
sorted_data = np.sort(data)  
ecdf = np.arange(1, len(data)+1) / len(data)
```



Helps in understanding:

- Data patterns and trends in real-world datasets
- Frequency and probability of occurrence of values
- Skewness, spread, and shape of data

Empirical Distribution (4)

- Presence of outliers and anomalies
- Underlying structure and variability in data

Relationship with Theoretical Distributions:

- Empirical distribution is data-driven, while theoretical distributions are model-driven.
- Common theoretical models include Normal, Poisson, and Exponential distributions.
- Empirical distribution is often used to test whether data fits a known model.
- Goodness-of-fit tests help compare empirical vs theoretical distributions.

Applications:

- Exploratory Data Analysis (EDA)
- Data preprocessing and cleaning
- Feature understanding in machine learning
- Detecting skewed or imbalanced data
- Decision-making based on real-world data patterns

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
 - Numeric Data Distributions
 - Categorical Data Distributions
- 3 Correlation Analysis
- 4 Statistical Significance

Numeric Data Distributions (1)

1. Histogram:

- A graphical representation of numerical data distribution.
- Divides data into bins (intervals) and shows frequency of each bin.
- It is an approximation of the underlying probability distribution.
- Useful for identifying shape, spread, skewness, and outliers.

Histogram Formula (Frequency Density):

$$d_i = \frac{f_i}{w_i}$$

where d_i is the frequency density of bin i , f_i is the frequency of bin i , and w_i is the width of bin i

Numeric Data Distributions (2)

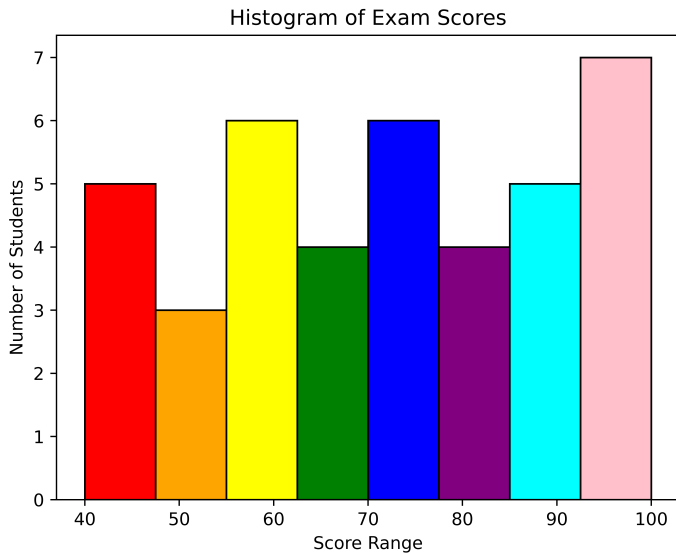
```
data = [
45, 50, 55, 60, 62, 65, 70, 72, 75, 78,
80, 82, 85, 88, 90, 92, 95, 96, 97, 98,
55, 60, 65, 70, 75, 80, 85, 96, 95, 100,
40, 42, 44, 46, 48, 52, 58, 63, 67, 73
]

counts, bins = np.histogram(data, bins=8)

# Colors for each bin
colors = ['red', 'orange', 'yellow', 'green', 'blue', 'purple', '
cyan', 'pink']

# Plot each bin as a bar with different color
plt.bar(bins[:-1], counts, width=np.diff(bins), align='edge',
color=colors, edgecolor='black')
```

Numeric Data Distributions (3)



Numeric Data Distributions (4)

2. Normal Distribution

A continuous probability distribution that is symmetric and bell-shaped. It is also known as the Gaussian distribution, named after Carl Friedrich Gauss. The graph of a normal distribution is perfectly symmetric about its center.

Key Characteristics

- Mean = Median = Mode i.e. all three measures of central tendency coincide at the center of the distribution.
- The distribution is completely defined by two parameters:
 - **Mean (μ)**: Determines the center (location) of the distribution.
 - **Standard Deviation (σ)**: Determines the spread (width) of the distribution.
- The curve approaches the x-axis but never touches it (asymptotic behavior).
- The total area under the curve is equal to 1, representing total probability.

Numeric Data Distributions (5)

Probability Density Function (PDF)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

μ = mean of the distribution

σ = standard deviation

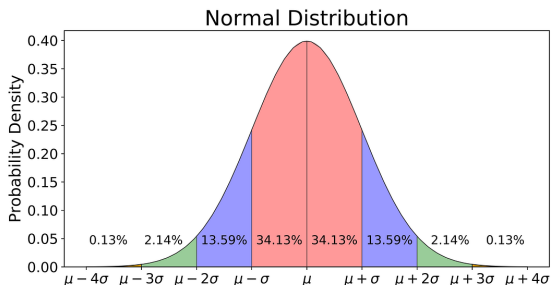
σ^2 = variance

Empirical Rule (68–95–99.7 Rule)

For a normal distribution:

- About 68% of data lies within 1 standard deviation of the mean ($\mu \pm \sigma$)
- About 95% lies within 2 standard deviations ($\mu \pm 2\sigma$)
- About 99.7% lies within 3 standard deviations ($\mu \pm 3\sigma$)

Numeric Data Distributions (6)



Standard Normal Distribution

A special case where:

- Mean $\mu = 0$
- Standard deviation $\sigma = 1$

Numeric Data Distributions (7)

Any normal variable X can be converted into a standard normal variable Z using:

$$Z = \frac{X - \mu}{\sigma}$$

This process is called standardization or finding the z-score.

Forms the basis of many statistical methods, including:

- Hypothesis testing
- Confidence intervals

Important Notes

- The normal distribution is an example of a continuous distribution, meaning probabilities are calculated over intervals rather than exact values.
- It plays a key role in the **Central Limit Theorem**, which explains why many real-world phenomena tend to follow a normal distribution when sample sizes are large.

Numeric Data Distributions (8)

```
from scipy.stats import norm

data = [
    45, 50, 55, 60, 62, 65, 70, 72, 75, 78,
    80, 82, 85, 88, 90, 92, 95, 96, 97, 98,
    55, 60, 65, 70, 75, 80, 85, 96, 95, 100,
    40, 42, 44, 46, 48, 52, 58, 63, 67, 73
]

data = np.array(data)

# Fit normal distribution
mu = np.mean(data)
sigma = np.std(data)

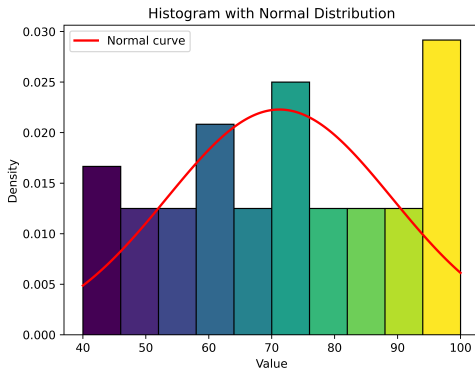
# Create histogram
counts, bins, patches = plt.hist(data, bins=10, density=True,
    edgecolor='black')

# Assign different colors to each bin
colors = plt.cm.viridis(np.linspace(0, 1, len(patches)))

for patch, color in zip(patches, colors):
    patch.set_facecolor(color)
```

Numeric Data Distributions (9)

```
# Normal curve
x = np.linspace(min(data), max(data), 200)
plt.plot(x, norm.pdf(x, mu, sigma), 'r', linewidth=2, label='
    Normal curve')
```



Numeric Data Distributions (10)

3. Exponential Distribution

- Models time between events in a Poisson process.
- Skewed distribution with rapid decay (right-skewed).
- Used in reliability analysis, queuing theory, survival analysis, and waiting-time problems.
- Describes how long we wait until the next event occurs.
- Has the **memoryless property**: future probability does not depend on past waiting time.
- Commonly used to model lifetimes of components with a constant failure rate.

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- λ = rate parameter (events per unit time)
- Mean (expected value): $\mathbb{E}[X] = \frac{1}{\lambda}$
- Variance: $\text{Var}(X) = \frac{1}{\lambda^2}$

Numeric Data Distributions (11)

- Cumulative distribution function (CDF):

$$F(x) = 1 - e^{-\lambda x}$$

- Survival function (tail probability):

$$P(X > x) = e^{-\lambda x}$$

Key Intuition:

- Small waiting times are most likely.
- Long waiting times are possible but become exponentially unlikely.
- Larger λ means more frequent events (shorter waiting times).
- Smaller λ means rare events (longer waiting times).

Memoryless Property:

- For $s, t \geq 0$:

$$P(X > s + t \mid X > s) = P(X > t)$$

- The process "resets" after every moment.

Numeric Data Distributions (12)

- The exponential distribution is the only continuous distribution with this property.

```
from scipy.stats import expon

data = [
    45, 50, 55, 60, 62, 65, 70, 72, 75, 78,
    80, 82, 85, 88, 90, 92, 95, 96, 97, 98,
    55, 60, 65, 70, 75, 80, 85, 96, 95, 100,
    40, 42, 44, 46, 48, 52, 58, 63, 67, 73
]

data = np.array(data)

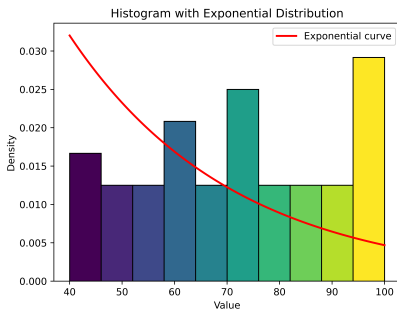
# Fit exponential distribution
# loc shifts the distribution; often forced to 0 for "pure"
# exponential
loc, scale = expon.fit(data)

# Create histogram
counts, bins, patches = plt.hist(data, bins=10, density=
    True, edgecolor='black')
```

Numeric Data Distributions (13)

```
# Assign different colors to each bin
colors = plt.cm.viridis(np.linspace(0, 1, len(patches)))
for patch, color in zip(patches, colors):
    patch.set_facecolor(color)

# Exponential curve
x = np.linspace(min(data), max(data), 200)
plt.plot(x, expon.pdf(x, loc, scale), 'r', linewidth=2,
        label='Exponential curve')
```



Numeric Data Distributions (14)

Example:

- Time between bus arrivals: $\lambda = 0.5$
- Average waiting time is 2 time units.
- Most buses arrive quickly, but some delays are long.
- Even if you have already waited 10 minutes, the expected additional waiting time remains the same.

4. Power Law Distribution (Pareto distribution)

- A few large values and many small values.
- Common in networks, social media, and economics.
- Often called a "heavy-tailed distribution" because extreme values occur more frequently than in exponential or normal distributions.
- No characteristic scale — the same pattern appears at different magnitudes (self-similarity).

$$P(x) = Cx^{-\alpha}$$

Numeric Data Distributions (15)

- α = scaling exponent (typically $\alpha > 1$ for real-world systems)
- C = normalization constant ensuring total probability sums to 1
- As x increases, probability decreases polynomially rather than exponentially

Key Properties:

- **Heavy tail:** Large events are rare but significantly more likely than in normal distributions.
- **Scale invariance:** If you rescale x , the functional form of the distribution remains similar.
- **Non-mean dominated:** In some cases, the mean is not representative due to extreme outliers.
- **Log-log linearity:** On a log-log plot, power laws appear as a straight line:

$$\log P(x) = \log C - \alpha \log x$$

Numeric Data Distributions (16)

```
# Parameters
alpha = 2.5
xmin = 1
size = 10000

# Generate power-law data
data = xmin * (1 + np.random.pareto(alpha - 1, size))

# Theoretical curve
x = np.linspace(xmin, max(data), 500)
C = (alpha - 1) * xmin**(alpha - 1)
pdf = C * x**(-alpha)

# Create side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

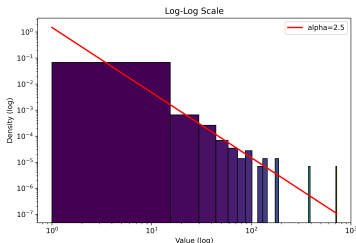
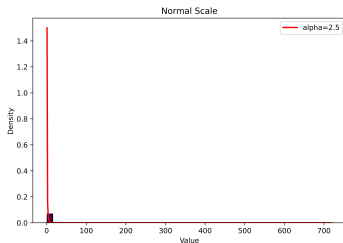
# ----- Plot 1: Normal scale -----
counts, bins, patches = axes[0].hist(data, bins=50, density
    =True, edgecolor='black')

# Color bins
colors = plt.cm.viridis(np.linspace(0, 1, len(patches)))
for patch, color in zip(patches, colors):
    patch.set_facecolor(color)
```

Numeric Data Distributions (17)

```
# ----- Plot 2: Log-Log scale -----
counts, bins, patches = axes[1].hist(data, bins=50, density
    =True, edgecolor='black')

# Color bins again
colors = plt.cm.viridis(np.linspace(0, 1, len(patches)))
for patch, color in zip(patches, colors):
    patch.set_facecolor(color)
```



Numeric Data Distributions (18)

Example:

- Website visits:
 - A small number of websites (e.g., Google, YouTube) receive billions of visits.
 - Most websites receive very few visits or remain almost unknown.
- City populations:
 - A few cities are extremely large (e.g., megacities).
 - Most cities and towns are relatively small.
- Social networks:
 - A few users have millions of followers (influencers/celebrities).
 - Most users have only a small number of connections.

Why it matters:

- Helps model real-world systems more accurately than normal distributions.
- Explains inequality and concentration effects in economics and society.
- Used in network science, internet traffic modeling, and risk analysis.

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
 - Numeric Data Distributions
 - Categorical Data Distributions
- 3 Correlation Analysis
- 4 Statistical Significance

Categorical Data: Bar Plots and Distributions (1)

1. Bar Plot

- Visual representation of categorical data.
- Displays frequency, count, or proportion of each category.
- Bars are separated by gaps to emphasize that categories are distinct (not continuous).
- The height (or length) of each bar is proportional to the value it represents.
- Can be displayed vertically or horizontally depending on readability.

Common Types of Bar Plots:

- **Simple bar chart:** shows one variable across categories.
- **Grouped bar chart:** compares multiple subgroups within each category.
- **Stacked bar chart:** shows composition of categories within a total.

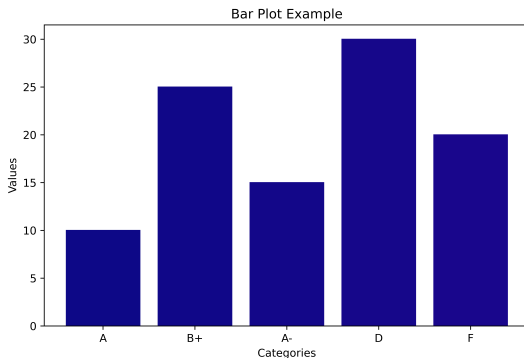
Categorical Data: Bar Plots and Distributions (2)

```
categories = ['A', 'B+', 'A-', 'D', 'F']
values = [10, 25, 15, 30, 20]

# Create bar plot
plt.figure(figsize=(8, 5))
bars = plt.bar(categories, values, edgecolor='black')

# Add colors
colors = plt.cm.plasma(range(len(bars)))
for bar, color in zip(bars, colors):
    bar.set_color(color)
```


Categorical Data: Bar Plots and Distributions (3)



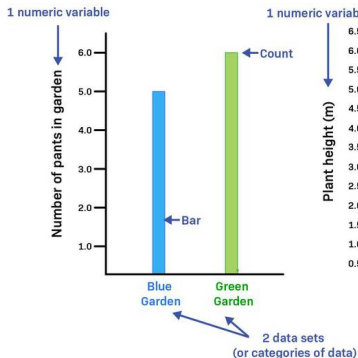
Key Features:

- Useful for comparing categories at a glance.
- Works well for both small and moderate numbers of categories.
- Not suitable for showing trends in continuous data (use line plots instead).

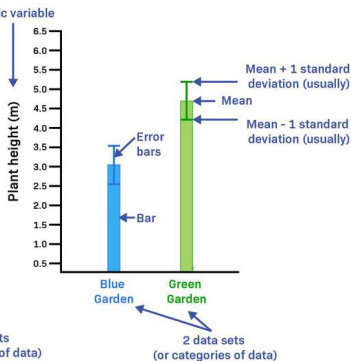
Categorical Data: Bar Plots and Distributions (4)

Example:

(A) Barplot representing amounts



(B) Barplot summarizing data sets



Categorical Data: Bar Plots and Distributions (5)

Binomial Distribution:

- Represents the number of successes in a fixed number of independent binary experiments.
- Each trial has only two outcomes: success/failure, yes/no, heads/tails.
- All trials are independent and have the same probability of success.
- Defined by two parameters: number of trials n and probability of success p .

Probability Mass Function (PMF):

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

- n = number of trials
- p = probability of success in each trial
- k = number of observed successes
- $\binom{n}{k}$ = number of ways to choose k successes from n trials

Key Properties:

Categorical Data: Bar Plots and Distributions (6)

- Mean (expected value): $\mathbb{E}[X] = np$
- Variance: $\text{Var}(X) = np(1 - p)$
- Distribution is symmetric when $p = 0.5$, otherwise it is skewed.
- Approaches a normal distribution for large n (Central Limit Theorem).

Key Assumptions:

- Fixed number of trials.
- Each trial is independent.
- Only two possible outcomes per trial.
- Constant probability of success across trials.

Example:

- Flipping a coin 10 times:
 - $n = 10, p = 0.5$
 - Random variable X = number of heads
- Other examples:
 - Number of defective items in a batch
 - Number of students passing an exam
 - Number of successful website clicks in n ads

Zipf's Law (1)

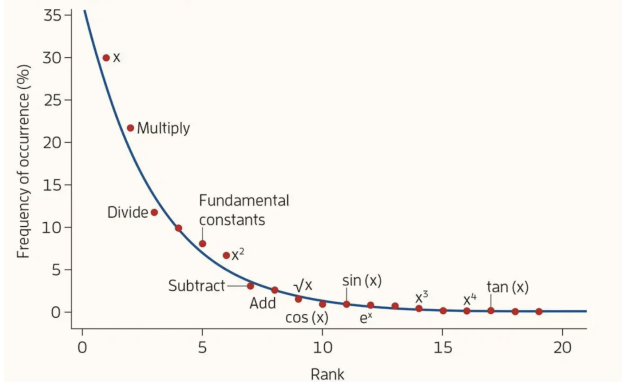
- **Frequency of an item is inversely proportional to its rank in a sorted list.**
- Commonly observed in language, text data, and many natural and social systems.
- A few items occur extremely frequently, while most occur rarely.
- Often considered a special case of power-law behavior.

Mathematical Form:

$$f(r) = \frac{C}{r^s}$$

- r = rank of the item (1 = most frequent)
- C = normalization constant
- $s \approx 1$ for many real-world datasets

Zipf's Law (2)



Key Intuition:

- The most frequent item appears roughly twice as often as the second most frequent, three times as often as the third, and so on (when $s \approx 1$).
- Produces a highly skewed distribution with a long tail.

Zipf's Law (3)

- Reflects imbalance in natural and human-generated systems.

Common Applications:

- **Language:** frequency of words in books, articles, and speech.
- **Web data:** popularity of websites or search queries.
- **City sizes:** population distribution across cities.
- **Social media:** frequency of hashtags, posts, or user interactions.

Example:

- In a large text corpus:
 - The word "the" appears extremely frequently.
 - Words like "data" or "model" appear less often.
 - Most words appear only a few times.
- Word frequency decreases rapidly as rank increases.

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
- 3 Correlation Analysis
 - Pearson Correlation
 - Cross-tabulation
 - Numeric and Categorical Correlation
- 4 Statistical Significance

Correlation Analysis (1)

- A statistical method to measure the relationship between variables.
- Helps identify strength and direction of associations.
- **Correlation $\neq$ Causation:** Just because two variables move together doesn't mean one causes the other. For example, ice cream sales and shark attacks are correlated (both increase in summer), but ice cream does not cause shark attacks.
- **Outliers:** A single extreme data point can significantly inflate or deflate the correlation coefficient.

Types of correlation analysis:

- **Numeric vs Numeric:**
 - **Pearson correlation** (linear relationship, parametric)
 - Spearman rank correlation (monotonic, non-parametric)
 - Kendall's tau (rank-based, robust to ties)
 - Partial correlation (controls for other variables)
 - Distance correlation (captures non-linear relationships)
- **Categorical vs Categorical:**
 - **Chi-square test of independence**

Correlation Analysis (2)

- Fisher's exact test (for small sample sizes)
- Cramér's V (effect size for Chi-square)
- Phi coefficient (for 2×2 contingency tables)
- Contingency coefficient
- Lambda coefficient (measures predictive association)
- **Numeric vs Categorical:**
 - Point-biserial correlation (binary categorical variable)
 - Biserial correlation (theoretical continuous split into binary)
 - Independent samples t-test (two groups)
 - Welch's t-test (two groups with unequal variances)
 - One-way ANOVA (three or more groups)
 - Welch's ANOVA (unequal variances across groups)
 - Mann-Whitney U test (non-parametric two-group comparison)
 - Kruskal-Wallis test (non-parametric ANOVA alternative)
 - Eta squared (η^2) (effect size for ANOVA)

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
- 3 Correlation Analysis
 - Pearson Correlation
 - Cross-tabulation
 - Numeric and Categorical Correlation
- 4 Statistical Significance

Pearson Correlation (1)

- Measures linear **relationship between two numeric variables**.
- Denoted by r .
- Range: $-1 \leq r \leq +1$
- Sensitive to linear relationships only (does not capture non-linear patterns).

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

- $\bar{x}, \bar{y}$ = sample means
- Measures how two variables vary together relative to their spread
- **Interpretation:**
 - $r = +1$: Perfect positive linear correlation
 - $r = -1$: Perfect negative linear correlation
 - $r = 0$: No linear relationship
 - Values closer to $|r| = 1$ indicate stronger linear association
- **Key Assumptions:**
 - Variables are continuous (numeric).

Pearson Correlation (2)

- Relationship is approximately linear.
- No extreme outliers (or results may be distorted).

Example: Given:

$$x = [28, 29, 29, 30, 31], \quad y = [132, 243, 180, 237, 219]$$

Step 1: Compute Means

$$\bar{x} = \frac{28 + 29 + 29 + 30 + 31}{5} = \frac{147}{5} = 29.4$$

$$\bar{y} = \frac{132 + 243 + 180 + 237 + 219}{5} = \frac{1011}{5} = 202.2$$

Step 2: Compute Deviations

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
28	132	-1.4	-70.2	98.28	1.96	4928.04
29	243	-0.4	40.8	-16.32	0.16	1664.64
29	180	-0.4	-22.2	8.88	0.16	492.84
30	237	0.6	34.8	20.88	0.36	1211.04
31	219	1.6	16.8	26.88	2.56	282.24

Pearson Correlation (3)

Step 3: Compute Sums

$$\sum (x - \bar{x})(y - \bar{y}) = 138.6$$

$$\sum (x - \bar{x})^2 = 5.2$$

$$\sum (y - \bar{y})^2 = 8578.8$$

Step 4: Apply Pearson Formula

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$r = \frac{138.6}{\sqrt{5.2 \times 8578.8}} \approx 0.656$$

```
import numpy as np
x = np.array([28, 29, 29, 30, 31])
y = np.array([132, 243, 180, 237, 219])
r = np.corrcoef(x, y)[1, 0]
print(r)
```

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
- 3 Correlation Analysis**
 - Pearson Correlation
 - Cross-tabulation**
 - Numeric and Categorical Correlation
- 4 Statistical Significance

Cross-tabulation: Correlation between Categorical Variables (1)

- Cross tabulation (or "crosstab") is a statistical tool used to analyze the relationship between categorical variables.
- Displays frequency distribution in a matrix (row vs column format) called **Contingency Table**.
- Each cell shows the count (or proportion) of observations for a category combination.
- It is the starting point for the **Chi-Square Test**, which determines if the relationship you see is statistically significant or just due to chance.

Group	Black Coffee	Latte	Total
Morning Person	40	10	50
Night Owl	15	35	50
Total	55	45	100

Table 1: Survey on Coffee Preference

- **Purpose:**

Cross-tabulation: Correlation between Categorical Variables (2)

- Identify patterns or associations between categories.
- Compare distributions across groups.
- Provide a basis for statistical testing of independence.
- **Statistical Link:**
 - Often used with the **Chi-square test of independence**.
 - Helps determine whether variables are statistically dependent or independent.
 - Measures association strength between categorical variables.
- **Interpretation:**
 - Large differences across rows/columns suggest possible association.
 - Similar distributions suggest weak or no relationship.

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
- 3 Correlation Analysis
 - Pearson Correlation
 - Cross-tabulation
 - Numeric and Categorical Correlation
- 4 Statistical Significance

Numeric and Categorical Correlation (1)

- **Mixed Variable Correlation:**

- Measures relationship between numeric and categorical variables.

- Common approaches:

- Group-wise comparison (mean/median analysis)
- ANOVA (Analysis of Variance)
- Point-biserial correlation (binary categories)

- **Example:**

- Salary (numeric) vs Job role (categorical)
- Age vs Customer segment

Table of Contents

1 Types of Variables: Numeric and Categorical

2 Empirical Distribution

3 Correlation Analysis

4 Statistical Significance

- p-value
- t-value
- Chi-Squared Test

Statistical Significance (1)

Statistical Significance:

- Determines whether observed results are likely due to chance or reflect a real effect in the population.
- Helps evaluate hypotheses in statistical inference.
- Provides a formal framework for decision-making using sample data.

Hypothesis Testing Framework:

- Null hypothesis (H_0): no effect or no relationship
- Alternative hypothesis (H_1): an effect or relationship exists
- Statistical tests evaluate evidence against H_0

Key Measures:

- **p-value:** Probability of observing results at least as extreme as the data, assuming H_0 is true
- **t-value:** Measures how far a sample statistic is from the null hypothesis in units of standard error
- **chi-squared statistic:** Measures discrepancy between observed and expected frequencies (categorical data)

Statistical Significance (2)

Decision Rule:

- If $p \leq \alpha$ (commonly 0.05), reject H_0
- If $p > \alpha$, fail to reject H_0

Interpretation:

- Statistically significant $\neq$ practically important
- Larger sample sizes can make very small effects statistically significant

Table of Contents

- 1 Types of Variables: Numeric and Categorical
- 2 Empirical Distribution
- 3 Correlation Analysis
- 4 Statistical Significance
 - p-value
 - t-value
 - Chi-Squared Test

p-value (1)

p-value

The **p-value** is the probability of obtaining results at least as extreme as the observed results, assuming the null hypothesis is true. The p-value (probability value) is a number that helps you determine if your data results are statistically significant or just a result of random chance.

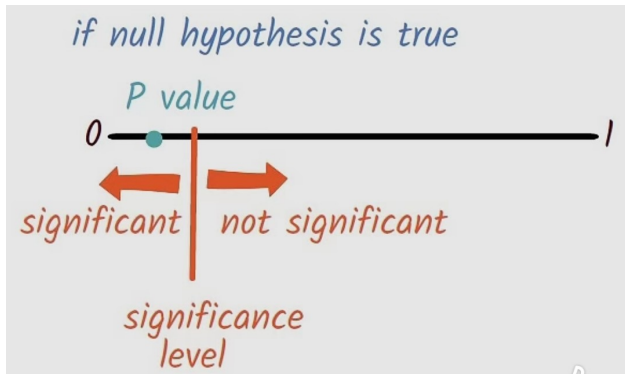
It helps determine whether an observed relationship (correlation or group difference) is statistically significant.

Interpretation:

- If $p \leq 0.05$: result is typically considered **statistically significant** (reject null hypothesis).
- If $p > 0.05$: result is **not statistically significant** (fail to reject null hypothesis).
- It answers the question: "If the 'null hypothesis' is true (meaning there is no real effect or relationship), how likely is it that I would see results at least as extreme as mine?"
- Probability of observing results under null hypothesis.

p-value (2)

- Smaller p-value indicates stronger evidence against null hypothesis.



p-value (3)

How to Read a P-Value?

- The p-value is a decimal between 0 and 1. The most common threshold for "significance" is 0.05.
- $P \leq 0.05$ (Small): You reject the null hypothesis. There is a strong chance that a real relationship exists. Your result is "statistically significant."
- $P > 0.05$ (Large): You fail to reject the null hypothesis. The relationship or difference you see is likely just "noise" or random coincidence.

p-value (4)

Example

Imagine you flip a coin 10 times and get 10 heads.

- **Null Hypothesis:** The coin is fair (50/50 chance).
- **P-Value:** The probability of getting 10 heads in a row with a fair coin is very low (about 0.001).
- **Decision:** Since 0.001 is much smaller than 0.05, you reject the idea that the coin is fair and conclude it's likely rigged.

A p-value of 0.05 does not mean there is a 95% chance that your hypothesis is correct. It only means that there is a 5% chance you would see these results if there was actually nothing happening at all.

Table of Contents

1 Types of Variables: Numeric and Categorical

2 Empirical Distribution

3 Correlation Analysis

4 Statistical Significance

- p-value
- **t-value**
- Chi-Squared Test

t-value (1)

t-value (t-statistic)

In statistics, the t-value (or t-statistic) is a number that measures how far your sample result is from the value stated in the null hypothesis, in terms of standard error.

- It answers the question: "**How many standard errors is my sample mean away from the hypothesized population mean?**"
- It is used in hypothesis testing when the population standard deviation is unknown and/or sample size is small.
- It helps determine whether the observed difference is likely due to a real effect or random chance.

t-value (2)

How to Interpret a t-value?

- The t-value can be positive or negative depending on the direction of difference.
- A t-value close to 0 means the sample result is very close to the null hypothesis.
- A large absolute t-value (positive or negative) indicates strong evidence against the null hypothesis.
- The t-value is compared with a critical value from the t-distribution to make a decision.

t-value (3)

Example

Imagine a class claims that the average study time is 5 hours per day.

- **Null Hypothesis:** The true mean study time is 5 hours.
- **Sample Result:** A group of students has an average of 6 hours with a small standard error.
- **t-value:** Suppose the calculated t-value is 3.2.
- **Decision:** Since 3.2 is large compared to the critical t-value (e.g., 2.0), we reject the null hypothesis and conclude that the average study time is significantly different from 5 hours.

The t-value does not directly give probability. It is used to compare against critical values or convert into a p-value to make a decision about the null hypothesis.

t-value (4)

t-value vs p-value Comparison

Feature	t-value (t-statistic)	p-value
Definition	A test statistic measuring distance between sample mean and null hypothesis in standard error units	A probability of observing the given result (or more extreme) if null hypothesis is true
Type	Numerical value (calculated statistic)	Probability value (0 to 1)
Purpose	Shows size and direction of difference	Shows strength of evidence against null hypothesis
Depends on	Sample data (mean, SD, size)	t-value and degrees of freedom
Decision use	Compared with critical t-value	Compared with significance level (e.g., 0.05)
Interpretation	Larger absolute value = stronger difference	Smaller value = stronger evidence against H_0

t-Tests (1)

A **t-test** is a statistical method used to determine whether there is a significant difference between the means of one or two groups. It is commonly used when the sample size is small and the population standard deviation is unknown.

Types of t-Tests

- 1. One-Sample t-Test:** Used to compare the mean of a single sample to a known value μ_0 .
- 2. Independent Two-Sample t-Test:** Used to compare means of two independent groups.
- 3. Paired t-Test:** Used when comparing means from the same group at different times.

When to Use a t-Test

A t-test is appropriate when:

- Comparing means of one or two groups
- Data is continuous (e.g., height, weight, scores)
- Sample size is small
- Data is approximately normally distributed

t-Tests (2)

- Observations are independent (except paired test)

When Not to Use a t-Test

Do not use a t-test when:

- Data is categorical
- Comparing more than two groups (use ANOVA)
- Data is highly skewed (use non-parametric tests)

A t-test helps determine whether observed differences between means are statistically significant or due to random chance.

t-Tests (3)

One-Sample t-Test

A one-sample t-test is used to compare the mean of a single sample to a known or hypothesized population mean.

When to use:

- Population standard deviation is unknown
- Sample size is small ($n < 30$) or data is approximately normal
- To test whether the sample mean differs from a known value

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Degrees of freedom: $df = n - 1$

Example

A class of 10 students has scores: 85, 88, 90, 92, 87, 84, 91, 89, 93, 86. Test if the class mean is significantly different from $\mu = 88$ at $\alpha = 0.05$.

Step 1: State Hypotheses

- Null hypothesis: $H_0 : \mu = 88$

t-Tests (4)

- Alternative hypothesis: $H_1 : \mu \neq 88$

Step 2: Calculate Sample Mean

$$\bar{x} = \frac{85 + 88 + 90 + 92 + 87 + 84 + 91 + 89 + 93 + 86}{10} = \frac{885}{10} = 88.5$$

Step 3: Calculate Sample Variance

$$\begin{aligned} s^2 &= \frac{\sum (x_i - \bar{x})^2}{n - 1} \\ &= \frac{(85 - 88.5)^2 + (88 - 88.5)^2 + \dots + (93 - 88.5)^2 + (86 - 88.5)^2}{9} \\ &= \frac{82.5}{9} \approx 9.17 \end{aligned}$$

Step 4: Calculate t-statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{88.5 - 88}{\sqrt{9.17}/\sqrt{10}} t \approx 0.52$$

t-Tests (5)

Step 5: Degrees of Freedom

$$df = n - 1 = 10 - 1 = 9$$

Step 6: Critical Value

At $\alpha = 0.05$ (two-tailed) and $df = 9$:

$$t_{\text{critical}} \approx 2.262$$

Step 7: Decision

$$|t| = 0.52 < 2.262$$

Therefore, we **fail to reject** the null hypothesis.

Step 8: Conclusion

There is **no statistically significant difference** between the class mean and $\mu = 88$ at the 0.05 significance level.

Table of Contents

1 Types of Variables: Numeric and Categorical

2 Empirical Distribution

3 Correlation Analysis

4 Statistical Significance

- p-value

- t-value

- Chi-Squared Test

Chi-Squared Test (1)

- Used to test relationship between categorical variables.
- Compares observed and expected frequencies.

Applications:

- Independence testing
- Goodness of fit

Example: Chi-Square Goodness-of-Fit Test

Exam Score = [45, 50, 55, 60, 62, 65, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95, 96, 97, 98, 55, 60, 65, 70, 75, 80, 85, 96, 95, 100, 40, 42, 44, 46, 48, 52, 58, 63, 67, 73]

1. Hypotheses

H_0 : The data is uniformly distributed across all bins

H_1 : The data is not uniformly distributed across all bins

2. Data and Binning

The data is grouped into the following class intervals:

Chi-Squared Test (2)

$(40-50), (50-60), (60-70), (70-80), (80-90), (90-100), (100-110)$

Total number of observations:

$$N = 40$$

Number of bins:

$$k = 7$$

3. Observed Frequencies

$$O = [6, 5, 7, 7, 6, 8, 1]$$

4. Expected Frequencies

$$E_i = \frac{N}{k} = \frac{40}{7} \approx 5.714$$

$$E = [5.714, 5.714, 5.714, 5.714, 5.714, 5.714, 5.714]$$

Chi-Squared Test (3)

5. Chi-Square Test Statistic

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

6. Calculation

Class Interval	O_i	E_i	$(O_i - E_i)$	$\frac{(O_i - E_i)^2}{E_i}$
40 – 50	6	5.714	0.286	0.014
50 – 60	5	5.714	-0.714	0.089
60 – 70	7	5.714	1.286	0.289
70 – 80	7	5.714	1.286	0.289
80 – 90	6	5.714	0.286	0.014
90 – 100	8	5.714	2.286	0.915
100 – 110	1	5.714	-4.714	3.889

7. Chi-Square Value

$$\chi_{\text{calc}}^2 = 5.498$$

Chi-Squared Test (4)

8. Degrees of Freedom

$$df = k - 1 = 7 - 1 = 6$$

9. Critical Value

At significance level $\alpha = 0.05$:

$$\chi_{0.05,6}^2 = 12.592$$

10. Conclusion

$$\chi_{\text{calculated}}^2 = 5.498 < 12.592$$

Therefore, we fail to reject H_0 .

There is no significant evidence against uniform distribution.

Chi-Squared Test (5)

```
import numpy as np
from scipy.stats import chisquare

data = np.array([
45, 50, 55, 60, 62, 65, 70, 72, 75, 78,
80, 82, 85, 88, 90, 92, 95, 96, 97, 98,
55, 60, 65, 70, 75, 80, 85, 96, 95, 100,
40, 42, 44, 46, 48, 52, 58, 63, 67, 73
])

# Step 1: Define bins
bins = [40, 50, 60, 70, 80, 90, 100, 110]

# Step 2: Observed frequencies
observed, _ = np.histogram(data, bins=bins)

# Step 3: Expected frequencies (uniform distribution assumption)
expected = np.ones_like(observed) * np.mean(observed)

# Step 4: Chi-square test
chi_stat, p_value = chisquare(f_obs=observed, f_exp=expected)

# Output
print("Observed frequencies:", observed)
```

Chi-Squared Test (6)

```
print("Expected frequencies:", expected)
print("Chi-square statistic:", chi_stat)
print("p-value:", p_value)

# Step 5: Decision
alpha = 0.05
if p_value < alpha:
    print("\nReject H0: Data is not uniformly distributed.")
else:
    print("\nFail to reject H0: No significant evidence against
          uniform distribution.")
```

Example (Cross Tabulation: Chi-Square Test for Independence)

A survey of 50 students records their preferred study method (Online, Offline) and their grade category (Pass, Fail):

	Pass	Fail	Total
Online	20	5	25
Offline	15	10	25
Total	35	15	50

Test if study method and grade category are in-

dependent at $\alpha = 0.05$. Also, specify which Python function(s) can be used.

Chi-Squared Test (7)

Step 1: State Hypotheses

- Null hypothesis: H_0 : Study method and grade category are independent
- Alternative hypothesis: H_1 : Study method and grade category are not independent

Step 2: Compute Expected Frequencies

$$E_{ij} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

$$E_{\text{Online, Pass}} = \frac{25 \times 35}{50} = 17.5$$

$$E_{\text{Online, Fail}} = \frac{25 \times 15}{50} = 7.5$$

$$E_{\text{Offline, Pass}} = \frac{25 \times 35}{50} = 17.5$$

$$E_{\text{Offline, Fail}} = \frac{25 \times 15}{50} = 7.5$$

Chi-Squared Test (8)

Step 3: Compute Chi-Square Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\chi^2 = \frac{(20 - 17.5)^2}{17.5} + \frac{(5 - 7.5)^2}{7.5} + \frac{(15 - 17.5)^2}{17.5} + \frac{(10 - 7.5)^2}{7.5} = 2.38$$

Step 4: Degrees of Freedom

$$df = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$$

Step 5: Critical Value

At $\alpha = 0.05$, $df = 1$:

$$\chi^2_{\text{critical}} = 3.841$$

Step 6: Decision

$$\chi^2 = 2.38 < 3.841$$

Fail to reject H_0 .

Chi-Squared Test (9)

Step 7: Conclusion

There is **no significant association** between study method and grade category.